Platform Development & Integration

CYBER THREATS & VULNERABILITIES 2 | ASSIGNMENT 5 | 09 NOVEMBER 2026

Executive summary

An isolated, Docker-hosted DeRexi Pulse Lab was built and exercised on Windows. The submitted evidence confirms a running, healthy container; 14 successful local runtime/security/dashboard checks; and a separately captured live regulatory-feed retrieval containing 87 normalized articles across four sources. This is a **validated Pulse Lab deployment**, not a claim that the full DeRexi governance application was containerized or production deployed.

Verified evidence at a glance

Docker container	Local checks	Live feed collection
Running / healthy	14 / 14 TRUE	87 normalized items
127.0.0.1:8080 only	Non-root, read-only root, no new privileges	4 / 4 feed sources report ok

Rubric-to-evidence alignment

Instructor requirement	Evidence supported	Scope / caveat
Docker environment installation and operation	Docker Desktop screenshot; security-summary.json records healthy runtime	Proves isolated Pulse Lab, not full DeRexi
Feed integration and management	container-feeds.json: 4 successful sources, counts and timestamps	A separate live run; dashboard is a synthetic fixture
Customized security configuration	Non-root UID, dropped caps, read-only root and localhost port	Configuration reported by container inspection
Analytics dashboard	Browser-based Pulse Lab: source bars, categories, item count and failure-state display; related checks pass	Synthetic dashboard; 1 displayed fixture item
Deployment documentation	Recorded container state, port binding, source evidence, test outcomes and procedures below	No full production runbook or live database tested

Evidence classification: **Executed and observed in an isolated environment.** The source JSON files and screenshot are the primary artifacts; historical Codex summaries are supplementary, not substitutes for raw evidence.

Technical implementation & verification

Architecture and data flow

Live-source validation path: Federal Register / FDIC / NIST / Federal Reserve public feed endpoints → collection and normalization → source classification and article metadata → *container-feeds.json*.

Isolated dashboard path: Synthetic fixture → Pulse Lab service inside Docker → dashboard at **127.0.0.1:8080**. It intentionally displays a simulated source failure state; it does not pull current publishers on that page.

Container configuration captured

Property	Observed value
Container	/derexi-ctv2-pulse-lab-1
Runtime state	running; healthy
Effective user	65532:65532 (non-root)
Root filesystem	Read-only = true
Linux capabilities	All dropped
Security options	no-new-privileges:true
Network exposure	127.0.0.1:8080 mapped to container 8080/tcp
Mounted volumes	0 mounts

Runtime and interface validation

The *container-checks.json* artifact records **14 of 14 checks as true**, covering: non-root process, capability dropping, prevention of privilege escalation, absence of database environment/module, read-only filesystem, health endpoint and content security policies, root page response, synthetic-data labeling, item count, source bars and simulated source failures.

Government feed retrieval (separate live check)

Source	Normalized items	Recorded status
Federal Register	38	OK
FDIC	25	OK
NIST	9	OK
Federal Reserve	15	OK
TOTAL	87	All four sources OK

Recorded retrieval timestamp: **2026-11-09T11:51:37Z** (UTC). The live artifact contains article IDs, source labels, published dates, URLs, categories and status metadata. Counts are collection output, not independently verified against each government publisher.

Evidence, procedures & boundaries

Operational verification steps

1. Start Docker Desktop with its WSL 2 backend and confirm the engine is running.
2. Confirm the Compose application named **derexi-ctv2** is running and its Pulse Lab service is healthy.
3. Open **http://127.0.0.1:8080** and confirm the synthetic-fixture label, source status bars, category and normalized-item count.
4. Inspect the three JSON evidence artifacts for the runtime hardening controls, 14 local checks and separately executed government feed results.
5. Preserve the container screenshot and raw logs; rerun isolated verification following changes without attaching real Supabase credentials.

Screenshot evidence - Docker Desktop

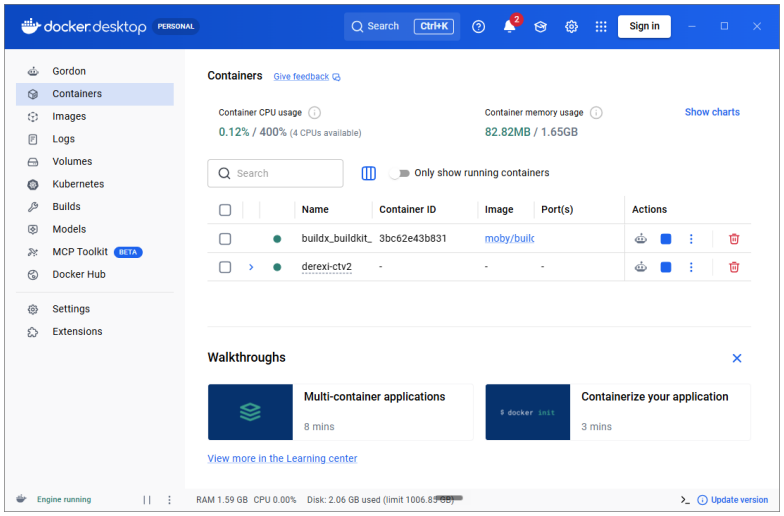


Figure 1. Docker Desktop shows the engine running and the **derexi-ctv2** Compose group with an active status indicator. Detailed healthy state is supported by the JSON inspection file.

Primary evidence index

Artifact	Purpose
security-summary.json	Container name, health, user, network and hardening settings
container-checks.json	14 Boolean security, health and synthetic-dashboard checks
container-feeds.json	87 normalized items, four source totals, URLs and UTC timestamps
Docker Desktop screenshot (09 Nov 2026)	Visual confirmation of running engine / Compose group

Limitations & completion verdict

Completed for the isolated Pulse Lab scope. The dashboard fixture is not the live feed display. The full DeRexi application, real user authentication, Supabase connectivity, persistence, and production deployment were not demonstrated in this evidence set. The source records support a scoped technical demonstration and should not be represented as enterprise-production certification.